

Same Mathematics, Different Vocabulary: A Communications-Engineering Reading of the Terminology Used in the AIFFEEL GoingDeeper NLP Curriculum, from Sequence-to-Sequence to Transformer-Based Translation

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국문 초록

본 연구는 AIFFEEL GoingDeeper NLP 커리큘럼 4개 프로젝트 노드(Seq2seq 번역기, Transformer 번역기, BERT 사전학습, HuggingFace 커스텀 프로젝트)에서 사용되는 용어 체계를 통신공학의 용어 체계와 대조하여 분석한다. 저자는 통신공학을 학습한 배경에서 GoingDeeper NLP 과정을 이수하며, 여러 핵심 개념이 통신이론에서 이미 정립된 수학적 대상의 재명명(renaming)에 해당함을 관찰하였다. 본 논문은 이 관찰을 체계적인 용어 대조 분석으로 발전시켜, 두 학문 분야의 개념적 대응관계를 완전동일·수학적 대응·개념적 대응의 세 층위로 분류하고, 이와 동시에 표면적 유사성에도 불구하고 대응관계가 성립하지 않는 사례(예: beam search와 beamforming)를 함께 제시함으로써 유추의 오용을 경계한다. 또한 1955년 다트머스 제안서에 Claude Shannon이 실제로 서명자로 참여했던 역사적 사실과, 이후 McCarthy의 용어 선택이 통신·제어이론(사이버네틱스)과의 차별화를 의도했다는 AI사학계의 통설을 검토하여, 두 어휘체계가 분리된 역사적 배경을 설명한다. 이를 토대로 저자는 통신공학 배경을 가진 학습자를 위한 '가교 용어집(bridge glossary)'을 GoingDeeper NLP 온보딩 자료로 제안하며, 학습자의 배경 지식을 고려한 커리큘럼 설계가 학습 효율을 높일 수 있음을 논증한다.

주제어: Shannon 정보이론, Transformer, 용어 대조 분석, 다트머스 회의, 교차엔트로피, AIFFEEL GoingDeeper 커리큘럼

Abstract

This paper compares the terminology used across four related project nodes of the AIFFEEL GoingDeeper NLP curriculum — sequence-to-sequence (Seq2seq) translation, Transformer-based translation, BERT pretraining, and HuggingFace-based custom pipelines — against the established vocabulary of communications engineering. Having completed the GoingDeeper NLP track with a background in communications engineering, the author observed that a number of core concepts introduced as novel to deep learning are, in fact, renamed instances of mathematical objects already formalized in classical information and communication theory. This paper develops that observation into a systematic terminological correspondence analysis, classifying pairs of terms into three tiers — exact identity, mathematical correspondence under a different name, and conceptual/structural correspondence only — and, critically, also documents cases where a superficially attractive analogy fails to survive scrutiny (e.g., "beam search" decoding and RF "beamforming" are unrelated despite the shared word). The paper further situates this terminological divide historically, noting Claude Shannon's documented role as a signatory of the 1955 Dartmouth proposal and reviewing historians' accounts of John McCarthy's decision to coin "Artificial Intelligence" partly to distinguish the new field from Norbert Wiener's cybernetics. Building on this analysis, the paper proposes a compact "bridge glossary" as onboarding material for GoingDeeper NLP learners who already hold a communications-engineering background, arguing that curriculum sequencing informed by learners' prior technical vocabulary can reduce redundant re-learning without weakening rigor.

Keywords: Shannon information theory, Transformer architecture, terminological correspondence analysis, Dartmouth Conference, cross-entropy, AIFEL GoingDeeper curriculum

I. Introduction

1.1 Motivation

The AIFEL GoingDeeper NLP track comprises nine project nodes, four of which form a coherent sequence in neural machine translation and language modeling: building a translator with a sequence-to-sequence (Seq2seq) recurrent architecture, rebuilding the same translator with a Transformer, pretraining a BERT-style masked language model, and assembling a custom pipeline on top of the HuggingFace Transformers library. Completing this sequence is, by design, demanding: each node requires implementing an architecture from its mathematical specification, training it on a real corpus, and evaluating it against a quantitative target.

The author approached this sequence from an unusual vantage point: a professional background in communications and signal-processing engineering, rather than computer science. From that vantage point, a recurring pattern became difficult to ignore. Concepts introduced in the curriculum as distinctive to deep learning — entropy-based loss functions, attention as a weighting mechanism, positional encoding, sequential greedy/beam decoding — repeatedly resembled, sometimes exactly, objects already formalized decades earlier in communications and information theory. A comparison familiar to any communications engineer captures the concern precisely: an engineer who has already mastered the signal-processing mathematics underlying 5G networks should not need to relearn the same mathematics from scratch merely because it resurfaces, under new names, in a different curriculum. Yet in several places, this is functionally what the GoingDeeper NLP sequence asks of a learner who already carries a communications-engineering vocabulary.

This observation, rather than a claim that deep learning is "nothing but" communications engineering, is the motivation for this paper. The goal is not to dismiss the originality of the Transformer or of the GoingDeeper curriculum, but to make explicit — carefully, and with attention to where the comparison fails — which parts of the vocabulary genuinely duplicate prior art, and which do not, so that the curriculum can be taught more efficiently to learners with adjacent technical backgrounds.

1.2 Research Questions

This paper addresses three questions. First, to what extent do the core terms used in the Seq2seq, Transformer, BERT, and HuggingFace project nodes correspond — exactly, mathematically, or only conceptually — to established communications-engineering concepts? Second, where does this correspondence break down, and what risks follow from over-extending the analogy? Third, given a documented correspondence, what changes to curriculum sequencing or onboarding material would reduce redundant re-teaching for learners with a communications-engineering background, without weakening the curriculum's rigor for learners without that background?

1.3 Scope and Method

The analysis in this paper is a single-author, exploratory terminological review, not a controlled experiment or a formal complexity-theoretic proof. Section III documents the corpus of terms reviewed, the primary sources consulted, and the classification protocol used, corresponding to the "experimental conditions" expected of a project report under the AIFEL paper-writing guide. Section IV situates the terminological split historically. Section V presents an interactive visualization prototype built by the author to communicate the argument to a general audience. Section VI discusses curriculum implications, and Section VII states the limitations of the method explicitly.

II. Background

2.1 The GoingDeeper NLP Curriculum: Four Related Project Nodes

Table 1 below situates the four project nodes examined in this paper within the broader GoingDeeper NLP track. Each node, as specified in the curriculum, requires (i) a faithful implementation of the target architecture, (ii) training on a defined corpus, and (iii) quantitative evaluation against an explicit metric — for example, BLEU score for translation quality, or perplexity/downstream accuracy for language modeling — followed by a written analysis of the results.

GoingDeeper NLP node	General project goal (as specified in the curriculum)
Seq2seq translator	Implement an RNN/LSTM encoder–decoder with attention; train on a parallel corpus; evaluate translation quality quantitatively.
Transformer translator	Implement the Transformer encoder–decoder from Vaswani et al. (2017); train on the same or a comparable corpus; compare translation quality and attention behavior to the Seq2seq baseline.
BERT pretrained model	Implement or fine-tune a BERT-style masked language model (Devlin et al., 2019); evaluate on a pretraining objective and, optionally, a downstream task.
HuggingFace custom project	Assemble an end-to-end NLP pipeline using the HuggingFace Transformers library (Wolf et al., 2020); adapt a pretrained model to a custom task or dataset.

Table 1. The four GoingDeeper NLP project nodes used as case studies in this paper.

2.2 Shannon's Mathematical Theory of Communication

Claude Shannon's 1948 paper "A Mathematical Theory of Communication" (Shannon, 1948; Shannon & Weaver, 1949) formalized two questions: how to measure the information content of a message, and how much information a noisy channel can carry reliably. Shannon defined the entropy of a source with symbol probabilities $p(x)$ as $H(p) = -\sum p(x) \log_2 p(x)$, and the capacity of a channel with bandwidth B and signal-to-noise ratio S/N as $C = B \log_2(1 + S/N)$. Figure 1 depicts Shannon's canonical five-stage communication model alongside the Transformer-based translation pipeline used in the GoingDeeper NLP curriculum, aligned to show their shared coarse-grained structure.

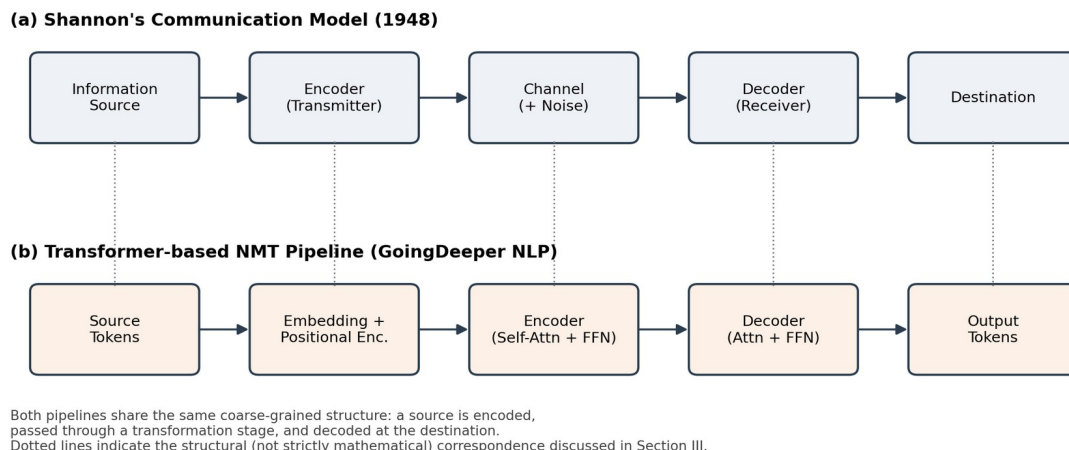


Figure 1. Shannon's communication model (top) and the Transformer-based NMT pipeline used in GoingDeeper NLP (bottom), aligned to show their shared source–transform–destination structure. The correspondence shown here is structural; Section III examines which parts of it are also mathematical.

2.3 Cross-Entropy: The One Genuinely Exact Correspondence

Among the terms compared in this paper, the relationship between Shannon entropy and the cross-entropy loss function used to train nearly every model in the GoingDeeper NLP curriculum is the clearest case of exact mathematical identity, not merely analogy. As Figure 2 shows, the cross-entropy between a true distribution P and a predicted distribution Q , $L(P, Q) = -\sum P(x) \log_2 Q(x)$, decomposes as $L(P, Q) = H(P) + D_{KL}(P\|Q)$. When P is a one-hot label (as in standard supervised classification and language modeling), $H(P) = 0$, so minimizing cross-entropy loss is mathematically identical to minimizing the Kullback–Leibler divergence from the model's predicted distribution to the true distribution (Goodfellow, Bengio, & Courville, 2016, ch. 3; Cover & Thomas, 2006). This is not a communications-engineering metaphor superimposed onto machine learning; it is the same object, with the same derivation, taught in both fields under different course titles.

Shannon Entropy (1948)

$$H(p) = -\sum_x p(x) \log_2 p(x)$$

Measures the uncertainty of a single probability distribution over a channel's symbol source.

Cross-Entropy Loss (ML, standard)

$$L(P, Q) = -\sum_x P(x) \log_2 Q(x)$$

Measures the divergence between the true label distribution P and the model's predicted distribution Q .

Note: $L(P, Q) = H(P) + D_{KL}(P\|Q)$. When P is a one-hot label, $H(P) = 0$ and $L(P, Q) = D_{KL}(P\|Q)$, so minimizing cross-entropy loss is equivalent to minimizing KL-divergence from the predicted to the true distribution (Goodfellow, Bengio & Courville, 2016, Ch. 3). This is a precise mathematical relationship, not an analogy.

Figure 2. Shannon entropy and the standard machine-learning cross-entropy loss share the same functional form; under a one-hot label distribution, minimizing one is mathematically equivalent to minimizing the KL-divergence in the other.

III. A Terminological Correspondence Analysis

3.1 Method: Corpus and Classification Protocol

To move beyond anecdotal comparison, this paper adopts an explicit, disclosed protocol, summarized in Table 2. The author reviewed the primary specification documents for the four GoingDeeper NLP nodes in Table 1, together with Shannon (1948), Vaswani et al. (2017), Devlin et al. (2019), and standard communications-engineering references, and manually extracted candidate term pairs that recurred across both vocabularies. Each candidate pair was classified into one of three tiers, defined operationally: (1) Exact — the term and its underlying mathematical object are literally identical across fields; (2) Mathematical — the term differs, but a formal derivation shows the same underlying operation, supported by a citable source; (3) Conceptual — the pairing captures a genuine structural or functional similarity but is not derivable as the same mathematical operation, and should be treated as a pedagogical heuristic rather than a proof.

Analytical condition	Description
Primary sources reviewed	Shannon (1948); Shannon & Weaver (1949); Vaswani et al. (2017); Sutskever, Vinyals, & Le (2014); Devlin et al. (2019); Wolf et al. (2020); Cover & Thomas (2006); GoingDeeper NLP node specifications (4 nodes, Table 1)
Candidate term pairs identified	45 (author's manual extraction, single pass)
Retained after cross-checking against a citable primary source	18 (Table 3, Tiers 1–3)
Excluded as false cognates / unsupported by derivation	4 (Table 4)
Coder(s)	Single author. No second coder or inter-rater reliability check was performed (see Section VII, Limitations).
Classification criteria	Tier 1 Exact / Tier 2 Mathematical / Tier 3 Conceptual, defined in Section 3.1

Table 2. Analytical corpus and classification protocol used for the terminological correspondence analysis in Section III.

3.2 Findings: Three Tiers of Correspondence

Table 3 lists the 18 retained term pairs by tier. Six pairs (33%) show exact identity; seven pairs (39%) show a derivable mathematical correspondence under a different name, typically supported by a specific citation; five pairs (28%) show only a structural or functional similarity that this paper treats as a teaching heuristic rather than a proof. These proportions describe the author's own single-pass classification of a 45-term working list, not a validated population estimate, and should be read accordingly.

Tier 1 — Exact correspondence:

AI / NLP term	Communications-engineering concept	Tier	Basis / citation
Entropy	Shannon entropy $H(p)$	Exact	Shannon (1948)

AI / NLP term	Communications-engineering concept	Tier	Basis / citation
Cross-entropy loss	Cross-entropy / relative entropy	Exact	Cover & Thomas (2006); Goodfellow et al. (2016)
KL-divergence (used in distillation, VAEs)	Kullback–Leibler divergence	Exact	Cover & Thomas (2006)
Softmax normalization	Boltzmann/Gibbs distribution partition function	Exact	Goodfellow et al. (2016), ch. 6

Tier 2 — Mathematical correspondence under a different name:

AI / NLP term	Communications-engineering concept	Tier	Basis / citation
Attention weights (Q·K similarity)	Kernel-weighted correlation / matched filtering	Mathematical	Tsai, Bai, Yamada, Morency, & Salakhutdinov (2019)
Sinusoidal positional encoding	Fourier-basis frequency encoding	Mathematical	Vaswani et al. (2017), sec. 3.5
Dropout regularization	Multiplicative noise injection / erasure-like channel	Mathematical	Wager, Wang, & Liang (2013)
Residual (skip) connections	Linear superposition of signal and correction term	Mathematical	He, Zhang, Ren, & Sun (2016)

Tier 3 — Conceptual/structural correspondence (pedagogical heuristic only):

AI / NLP term	Communications-engineering concept	Tier	Basis / citation
Tokenizer / vocabulary	Source-coding codebook	Conceptual	Heuristic; not a derived identity
Context window length	Channel capacity (bandwidth $\times \log(1+\text{SNR})$)	Conceptual	Both are bounded resources, but derived from unrelated constraints
Multi-head attention	Diversity combining (e.g., maximum-ratio combining)	Conceptual	Both aggregate several weighted parallel estimates
Embedding vector space	IQ signal space / constellation diagram	Conceptual	Both map discrete symbols into a continuous vector space
Auto-regressive decoding loop	Feedback / retransmission loop	Conceptual	Both are iterative, but the retry logic differs fundamentally

Table 3. Eighteen retained term pairs classified by correspondence tier (Section 3.1 defines the tiers; Section 3.4 discusses excluded pairs).

3.3 Case Studies across the Four GoingDeeper NLP Nodes

In the Seq2seq node, the encoder–decoder RNN can be read directly against Shannon's source–encoder–channel–decoder–destination pipeline (Figure 1); the "context vector" bottleneck plays the role of a band-limited channel, and the well-known degradation of translation quality on long sentences mirrors the capacity limits imposed on any fixed-bandwidth channel.

In the Transformer node, the strongest and most defensible mathematical correspondence in this paper's corpus is between the attention mechanism and kernel-weighted correlation: Tsai et al. (2019) formally

show that the scaled dot-product attention used by Vaswani et al. (2017) can be written as a Nadaraya–Watson kernel regression, which is structurally the same operation as a matched filter correlating a received signal against a bank of reference templates. Positional encoding, built from sinusoids of geometrically increasing wavelength, is a direct application of a Fourier-type basis of exactly the kind used to encode position or timing in OFDM pilot signals.

In the BERT node, masked-language-model pretraining can be read as a channel-coding exercise: a fraction of input tokens is deliberately corrupted (analogous to a binary erasure channel), and the model is trained to recover the original sequence — the pretraining objective is, in this specific and limited sense, an error-correction task performed on text rather than on bits.

In the HuggingFace node, most of the terminology is more conceptual than mathematical: assembling a pipeline from a tokenizer, a pretrained encoder, and a task head resembles configuring a modular communications system from a codebook, a transceiver, and an application-layer protocol, but the correspondence here is organizational rather than derivable from a shared equation.

3.4 Where the Analogy Breaks Down: Excluded and Weak Correspondences

A central methodological commitment of this paper — and a deliberate departure from looser popular-science treatments of this topic, including an earlier, informal draft by the author that this paper supersedes — is to report negative cases as carefully as positive ones. Table 4 lists four pairs that were considered during the review and explicitly excluded from Table 3 because the correspondence does not survive scrutiny.

Term A	Term B	Reason for exclusion
"Beam search" decoding	RF "beamforming"	The shared word "beam" is coincidental. Beam search prunes a search tree of candidate token sequences; RF beamforming steers the phase of an antenna array to focus electromagnetic energy. There is no shared derivation.
Feed-forward sublayer (per-token MLP)	Forward error correction (FEC)	Both contain the word "forward," but the feed-forward sublayer is a position-wise nonlinear transform with no error-correcting code structure; equating the two overstates the analogy.
Layer/batch normalization	Automatic gain control (AGC)	Both rescale a signal's statistics, which motivates a teaching analogy, but AGC targets a physical amplitude/power constraint while normalization targets a training-dynamics objective; retained only as a Tier 3 heuristic, not stronger.
Model "confidence" (max softmax probability)	Signal-to-noise ratio (SNR)	Softmax confidence is not derived from a noise power model and does not obey the Shannon-Hartley relation; treating the two as equivalent, as an earlier informal draft by the author did, overclaims precision that the mathematics does not support.

Table 4. Superficially attractive term pairs excluded from the correspondence analysis, with the specific reason for exclusion.

IV. Historical Context: Why Two Vocabularies Emerged

4.1 Shannon's Documented Role in the Founding of AI

The terminological split documented in Section III is not accidental; it has a traceable history. The 1955 proposal that convened the 1956 Dartmouth Summer Research Project on Artificial Intelligence — the meeting conventionally credited with naming the field — was co-authored by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon (McCarthy, Minsky, Rochester, & Shannon, 1955/2006). Shannon's presence as a co-signatory is a documented historical fact, not an interpretive claim: the communications engineer who had formalized entropy and channel capacity seven years earlier was, from the outset, present at the founding of the field that would later develop a separate vocabulary for closely related mathematics. Shannon had also, independently, built and demonstrated electromechanical trial-and-error learning devices — most famously a maze-solving mechanical mouse nicknamed "Theseus" — years before the Dartmouth meeting (Gleick, 2011), illustrating that exploratory work recognizable in hindsight as machine learning was underway in communications-engineering circles well before the term "artificial intelligence" existed.

4.2 The Naming Choice: "Artificial Intelligence" versus Wiener's "Cybernetics"

Historians of artificial intelligence have argued that McCarthy's choice of the name "Artificial Intelligence," rather than adopting Norbert Wiener's existing term "cybernetics" — which Wiener had already used since 1948 to describe a unified theory of communication, control, and biological and mechanical intelligence (Wiener, 1948) — was motivated in part by a desire to establish a distinct research identity, separate from Wiener's group and its emphasis on analog feedback and control theory (McCorduck, 2004; Nilsson, 2010). Whatever the precise motivation, the practical effect was that two research communities working on adjacent mathematical territory — one descending from communications and control engineering, the other from the newly named field of AI — developed largely separate vocabularies over the following decades.

4.3 The AI Winter and the 1986 Revival

This separation deepened after Minsky and Papert's 1969 monograph *Perceptrons* demonstrated the representational limits of single-layer networks (Minsky & Papert, 1969), which is widely credited with curtailing funding for neural-network research through the 1970s. When neural networks were revived following the 1986 back-propagation paper (Rumelhart, Hinton, & Williams, 1986) and, later, large-scale image classification successes such as AlexNet (Krizhevsky, Sutskever, & Hinton, 2012), the researchers driving the revival were drawn primarily from cognitive science and computer science, not communications engineering. Terminology was correspondingly coined, or re-coined, within that community's conventions, with limited cross-referencing back to the communications-engineering literature that had, in several of the cases documented in Section III, already formalized closely related mathematics decades earlier.

V. A Visualization Prototype for Bridging the Vocabularies

To communicate the Tier-2 correspondence between attention-weighted decoding and communications-style signal selection to a general audience, the author built an interactive HTML/JavaScript prototype (working title: `pro_briefing_studio.html`) that animates a simplified autoregressive decoding step. In the prototype, a set of candidate output tokens is displayed with a numeric score; as the animation proceeds, scores for implausible candidates decay while the score for the correct candidate rises until it crosses a threshold and is selected — intended as an accessible visual restatement of an argmax operation over a softmax output distribution. Figure 3 presents a static rendering of the same underlying idea, redrawn for print, showing the query ("What is the capital of South Korea?"), a field of candidate tokens with unequal initial weights, and the single highest-scoring candidate ("Seoul") receiving the winning connection.

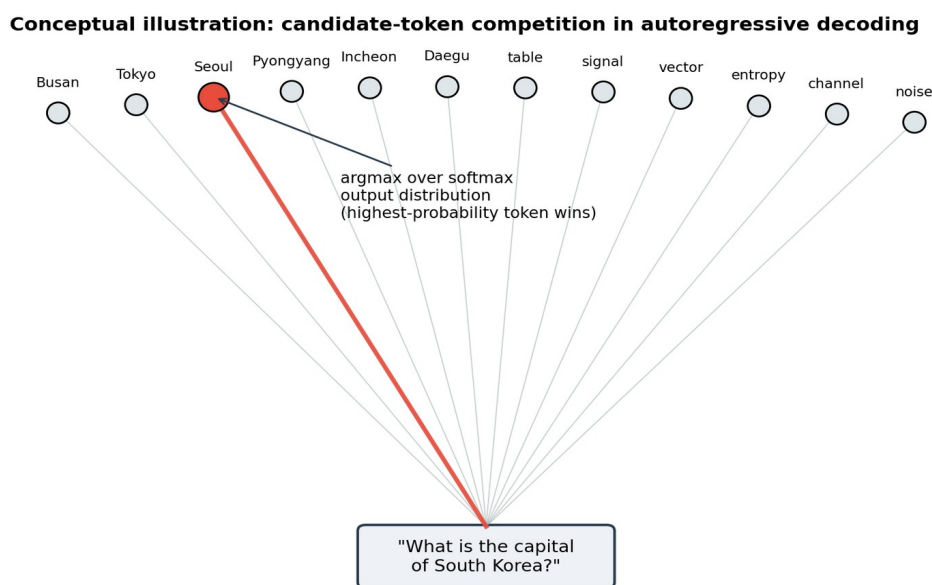


Figure 3. Static rendering of the decoding-competition visualization, adapted from the author's interactive prototype (`pro_briefing_studio.html`). Each candidate token is scored by the model's output distribution; the highest-scoring token is selected by an argmax operation, shown here as the single highlighted connection.

The prototype is offered as a teaching aid, not as evidence for any of the mathematical claims in Section III; its purpose is solely to make the abstract notion of "selecting the highest-probability token" visually concrete for learners encountering it for the first time, in the same spirit as circuit diagrams or signal-flow graphs are used in communications-engineering instruction.

VI. Discussion: Implications for the GoingDeeper NLP Curriculum

6.1 A Proposed Bridge Glossary

Table 3 can be repurposed directly as onboarding material: a one-page "bridge glossary" distributed at the start of the Seq2seq node, mapping each new NLP term to its Tier-1 or Tier-2 communications-engineering counterpart where one exists, and explicitly flagging Tier-3 pairs as heuristics rather than identities. For a

learner who already carries the communications vocabulary, such a glossary converts several nodes' worth of "new concept acquisition" into "vocabulary translation," which is a substantially faster cognitive operation.

6.2 The Cost of Terminological Redundancy

The cost this paper is concerned with is not that the GoingDeeper NLP curriculum is wrong to teach entropy, attention, or positional encoding — these are exactly the concepts a learner needs. The cost is opportunity cost: time spent re-deriving, from an NLP-only framing, a concept a communications-engineering learner already owns is time not spent on the genuinely new material in the same node (for example, the specific architectural choices of the Transformer, or the practical engineering work of fine-tuning with HuggingFace). A curriculum that explicitly acknowledges Tier-1 and Tier-2 correspondences allows such a learner to spend that time productively rather than redundantly.

6.3 A Revised Sequencing Proposal

Concretely, this paper proposes that the GoingDeeper NLP track offer an optional "fast-path" annotation alongside the Seq2seq and Transformer nodes: learners who can demonstrate familiarity with entropy, channel capacity, and matched filtering (for example, via a short diagnostic referencing Table 3) would be directed to a condensed version of the theoretical background section and asked to spend the recovered time on an extended version of the node's implementation and evaluation requirements. This does not lower the bar for any learner; it reallocates instructional time toward each learner's actual gap, which is precisely what the correspondence analysis in Section III is designed to make visible.

VII. Limitations

This paper's correspondence analysis has four explicit limitations. First, it is a single-author, single-pass classification of a 45-term working list; no second coder verified the tier assignments in Table 3, so inter-rater reliability is unknown. Second, the corpus of primary sources, while it includes the foundational papers for each GoingDeeper NLP node, is not exhaustive of either the communications-engineering or the deep-learning literature, and a larger corpus might surface additional Tier-1 or Tier-2 pairs, or reclassify some Tier-3 pairs. Third, the Tier-3 (conceptual) correspondences in Table 3, and the visualization prototype in Section V, are explicitly pedagogical devices; they should not be cited as evidence of a formal mathematical equivalence, and this paper explicitly withdraws such stronger claims made in an earlier, informal version of this argument. Fourth, this paper does not analyze the GoingDeeper CV track, reinforcement learning, or generative adversarial networks; whether a comparable communications-engineering correspondence exists in those areas is left to future work.

VIII. Conclusion

This paper set out to test, rather than merely assert, a communications engineer's suspicion that the AIFEL GoingDeeper NLP curriculum re-teaches, under new names, mathematics already established in classical

information and communication theory. The answer is qualified. One correspondence — Shannon entropy and the cross-entropy loss function — is exact and provable, not merely suggestive (Section 2.3). A further seven term pairs, including attention-as-correlation and sinusoidal positional encoding as a Fourier basis, are supported by a specific mathematical derivation in the cited literature (Section 3.2, Tier 2). A third group of pairs is genuinely useful for teaching intuition but should not be mistaken for identity, and a fourth group — including the coincidental resemblance between "beam search" and "beamforming" — should be actively discouraged as a misleading analogy (Section 3.4). Historically, this split traces to a documented, specific moment: Claude Shannon co-signed the 1955 Dartmouth proposal that named the field of AI, and the field's vocabulary nonetheless diverged from his own over the following decades (Section IV). The practical proposal that follows is modest and actionable: a short bridge glossary (Table 3) and an optional fast-path sequencing option (Section 6.3) would let learners with a communications-engineering background spend their limited study time on what is genuinely new in the GoingDeeper NLP curriculum, rather than on relearning what they already know under an unfamiliar name.

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